



Lift Regulation Using Closed-Loop Feedback and Control

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Abstract

A closed-loop feedback control system tuned by a PID controller was designed and built to achieve a target lift coefficient in both steady state as well as unsteady lift coefficient variation at reduced frequencies ranging from 0.05 to 0.3 at the University of Dayton Water Tunnel. Through the integration of a dSPACE MicroLabBox with a linear actuator and a force sensor, a semi-span AR 2.5 NACA 0015 wing was actuated using closed loop control to achieve a desired lift coefficient condition. The control framework actively senses the instantaneous lift on a wing using a force transducer and commands the changes in the instantaneous angle of attack of the wing to achieve the target lift coefficient. Since there is no plant model, the control system has no knowledge of aerodynamics and operates purely through sensor feedback. The target lift coefficient is then compared with the Theodorsen's model to determine whether the angle of attack input to the model matches with that of the angle of attack variation sensed by the actuator to achieve the desired lift coefficient. The long term goal of this study is to create a closed loop real time controller framework which can be used to leverage the controls methodologies that have been the result of years of experiments in open-loop analysis of lift mitigation from gusts and other disturbances.

Nomenclature

α	= geometric angle of attack (radians)	$\ddot{h}$	= plunge acceleration (m/s ²)
α_d	= desired geometric angle of attack (rad)	k	= reduced frequency
c	= chord	ω	= angular frequency (rad/s)
C_L	= lift force coefficient	f	= frequency (Hz)
s	= wing positing with respect to gust	t	= time (s)
U_∞	= freestream velocity (m/s)	τ	= convective time
$\dot{\alpha}$	= pitch rate (rad/s)	A	= C_L amplitude
$\ddot{\alpha}$	= pitch acceleration (rad/s ²)	Abbreviations	
b	= semi-chord (m)	GR	= Gust Ratio
a	= pitch axis location in semi-chord (m)	PID	= Proportional Integral Derivative
$C(k)$	= Theodorsen's function	LPF	= Low Pass Filter
$\dot{h}$	= plunge rate (m/s)		

I. Introduction

Much work has been done in the fluid dynamics community to understand the effects of gusts on airfoils and wings. The goal of many of these studies is to fit the experimental results with gust interactions to well-known models for the forces on an airfoil or wing when a gust is encountered. The gust interactions with the wings are usually studied by impinging an airfoil or a wing with a well-defined gust profile. A numerical/algebraic model is then derived to model the wing/airfoil response to the gust or at least a portion of the wing/airfoil response to the gust. However, successful modeling of a wing's response to different types of gusts (streamwise, transverse, vortical, etc.) has remained a velleity of many years, especially at high effective angles of attack.

In a study performed by Angulo et al. [1], two gust profiles were used in the transverse direction, a sine squared profile and a top hat profile, to affect a flat plate being towed through a water tank. Küssner model was used to predict the plate's response for both of these gust profiles. In case of the top-hat gust, the Küssner model

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underpredicted the lift force at higher gust ratios and in the case of the sine square profile, the model overpredicted the lift force over the range of gust ratios except between the gust ratio of 2 to 3 as shown in Figure 1. The deviations from the Küssner model was attributed to the vorticity shed from the wing and the vorticity present in the gust itself which is referred to as gust force. It is clear that the wing's response to gusts greatly differ depending on the type of gust profile. While understanding the aerodynamics is crucial in successfully modeling the wing's response, a closed loop control might be a simpler alternative for gust mitigation. The plant model in the closed loop control can be trained by the results from experimental data set or can be designed to work without any knowledge of the plant model as well using a well-tuned PID controller which can later be used to develop robust plant models.

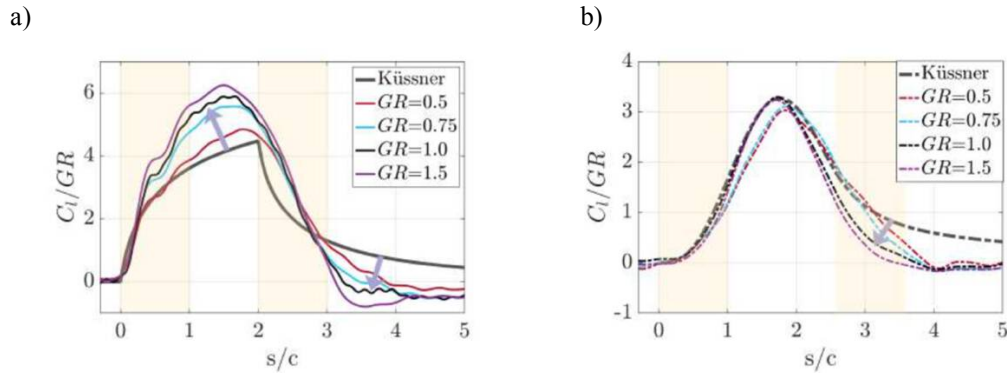


Figure 1 Comparison of experimental data to Küssner model for a) top hat and b) sine squared gusts [1]

Sedky et al. [2] showed the advantages of using closed loop control in gust mitigation. Similar to Andreu Angulo et al. [1], Sedky et al. [2] performed a transverse gust study using sinusoidal gust profile. When the wing encounters a sinusoidal gust profile, it is subjected to a non-uniform velocity field which can be decomposed as a sum of virtual angle of attack and a virtual camber. Using this decomposition, a modified Goman–Khrabrov Lift Model was used to derive an expression for the effective angle of attack. Using the modified GK model, a closed loop control simulation and an experimental open loop control was employed for gust mitigation. The results from the closed loop control simulation study were quite impressive for the period when the test article was encountering the gust. However, the experimental open loop control failed to mitigate the effects of lift after the gust encounter. The coefficient of lift time histories at different angles of attack and gust ratios for this experiment are shown in Figure 2. This deviation was accredited to the inability of the model to capture the effects of the leading and trailing edge vortices. This study pointed to the need for implementing a closed loop control for effective gust mitigation and to characterize the system response.

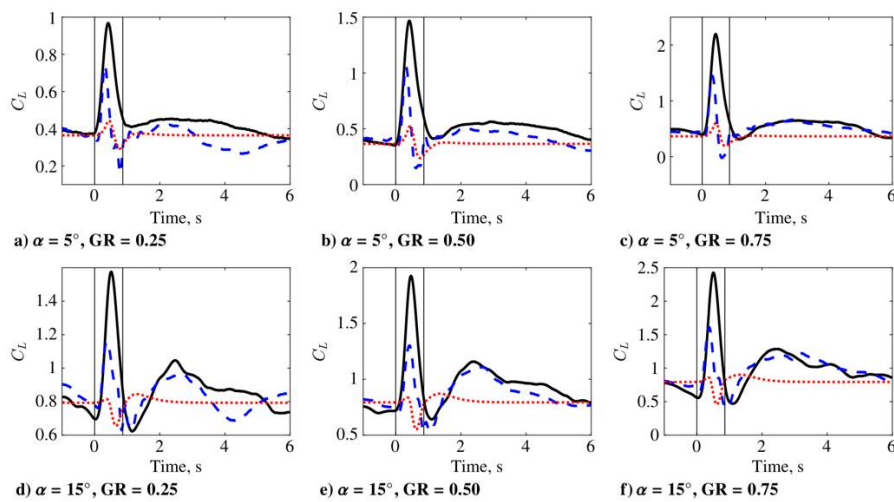


Figure 2 Comparison of experimental C_L time history without open loop control (—) and with open loop control (---), to closed-loop control simulation (---) [2]

Closed loop control is being used to alleviate gust control on small scale UAVs. Yeo et al. [3] developed a closed loop control for a small, multi-mode aircraft that operates in both helicopter and fixed-wing mode to control the vehicle in gusty environments and during aggressive maneuvers. A nonlinear controller using onboard flow measurements to feedback-linearize the system was experimentally demonstrated to mitigate the effect of vertical gusts on pitch and heave motions. Most quadrotor type UAVs are inherently unstable and more often a PID controller is needed to stabilize the altitude and the Euler angles of the system [4]. Murua et al. [5] simulated the gust load alleviation and trajectory regulation of HALE (High Altitude Long Endurance) flexible aircraft using PID control. A simple PID has been shown to perform well in substantially bringing down the bending strains and providing the necessary guidance authority. Asgharnia et al. [6] simulated the use of RBF based fractional-order PID (FOPID) to control the pitch angle of a 5-MW wind turbine blade to mitigate mechanical loads.

Previous work by Kerstens et al. [7] showed that a wing outfitted with active flow control devices undergoing streamwise velocity variations could mitigate the lift changes at reduced frequencies below 0.09. The study showed that without using a known unsteady aerodynamic model and instead using black-box system identification for the flow control devices and their response to gusting, they could achieve a practical model for the dynamics of the system. This process involved inputting pseudo-random binary signal step inputs to the system and measuring the system response. This data is then used to fit a transfer function for use in designing the controller. Similar system identification was done to motor controller used in the present study.

This main objective of the work presented in this study is to develop a closed-loop control framework necessary for performing gust mitigation studies at the University of Dayton Water Tunnel. The efficacy of the closed-loop control system is measured by the ability of the control system to achieve a target coefficient of lift in both static and dynamic conditions. The desired dynamic variation of coefficient of lift was then matched with the coefficient of lift variation from Theodorsen's model to determine if the corresponding angle of attack variation from the model matches with that of the angle of attack variation achieved by the controller.

Theodorsen's model gives a solution to the unsteady airloads on a 2-D harmonically oscillated airfoil in inviscid, incompressible flow, and subject to small disturbance assumptions. In coefficient form, Theodorsen's model can be expressed as,

$$C_l = \pi b \left(\frac{\dot{\alpha}}{U_\infty} + \frac{\ddot{h}}{U_\infty^2} - \frac{ba\ddot{\alpha}}{U_\infty^2} \right) + 2\pi C(k) \left[\frac{\dot{h}}{U_\infty} + \alpha + b \left(\frac{1}{2} - a \right) \frac{\dot{\alpha}}{U_\infty} \right] \quad (1)$$

Where $C(k)$ is a complex valued transfer function known as Theodorsen's function, which accounts for the effects of the shed wake on the unsteady airloads. The function is given by,

$$C(k) = F(k) + iG(k) \quad (2)$$

where $F(k)$ and $G(k)$ are shown in Figure 3. The first set of terms in Equation 1 represents the added mass and the second set of the terms represents the creation of circulation about the airfoil.

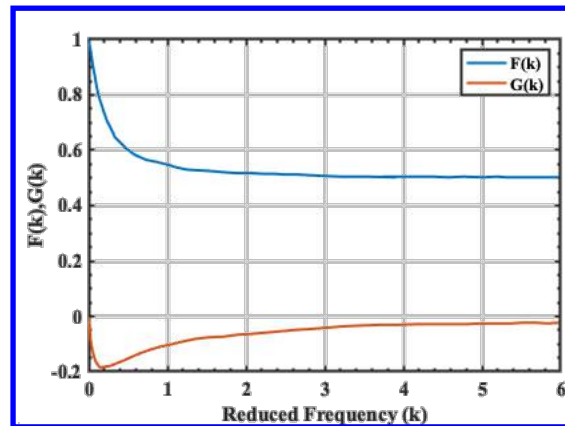


Figure 3 Variation of Theodorsen's function with reduced frequency

For pure pitch oscillations, the model reduces to Equation 3 [8]

$$C_l = 2\pi (F[1 + ik] + G[i - k])\alpha e^{i\omega t} + \pi k \left(i - \frac{k}{2}\right) \alpha e^{i\omega t} \quad (3)$$

The variation of coefficient of lift for a given angle of attack input modeled from Equation 3 is shown in Figure 4.

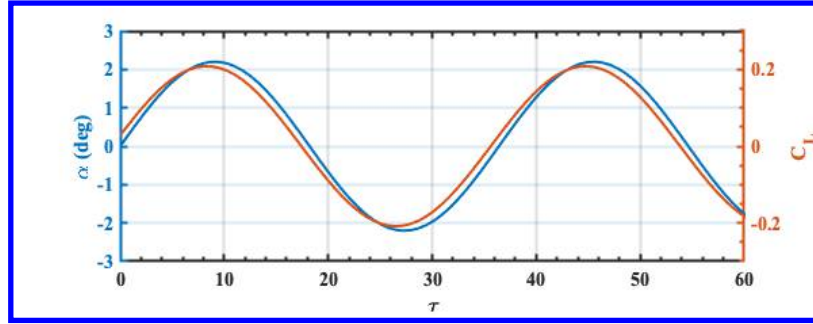


Figure 4 Variation of coefficient of lift for the given angle of attack input given by Theodorsen's model

It can be seen that for a given angle of attack variation, the Theodorsen model shows a variation in lift coefficient with a phase shift. In the present study, the given input lift coefficient variation is compared with the Theodorsen model to determine if the angle of attack variation imposed by the model matches that of the angle of attack variation sensed by the closed loop controller.

II. Experimental Setup

All experiments were performed at the University of Dayton Water Tunnel, shown in Figure 5a. The facility is a horizontal free-surface water tunnel with a 6:1 contraction and a 38 cm wide and 46 cm high test section of 152 cm length. The free-surface speed range is 7.5 cm/s to 38 cm/s. For this study, the freestream velocity of the tunnel was set to 0.371 m/s and a NACA0015 wing with a 4-inch (0.1016m) chord and 10-inch (0.254m) span was used. The tunnel is equipped with a one degree of freedom motion stage consisting of a H2W Technologies linear motor driven by a Xenus XTL-230-18 motor driver from Copley Controls. This motor is outfitted with a rack and pinion gear that converts its linear motion to a rotational motion in order to pitch a test article in the tunnel test section. The closed-loop feedback and control for this system is performed with a dSPACE MicroLabBox with BNC type terminations using the MATLAB LabVIEW Real Time Interface (RTI) programming environment. The MicroLabBox is capable of up to 24 channels of analog input and 16 channels of output in 16-bit at 1 Msps. There are also 48 channels of bidirectional digital inputs and outputs available. The MicroLabBox was ran at 10khz for these experiments although it is capable of up to 2Ghz processing in real-time. Force measurements were performed with an ATI Nano25 force transducer with NET F/T box interfaced with the dSPACE equipment using the ethernet protocol blocks in Simulink. A wiring schematic of the feedback system is shown in Figure 5c. The force transducer F/T outputs are taken into the MicroLabBox and used to calculate forces on the test article. The motor driver is connected to two analog outputs, one for enabling the motor and the other for commanding the current of the motor. Another connection between the motor driver and MicroLabBox is made via the digital I/O ports to receive information from the motor encoder.

The goal of this experimental setup is to create a system that can employ different aerodynamic controllers for closed-loop feedback in unsteady aerodynamic applications. Given this goal, the system design consists of two individual feedback controllers. The first is a state feedback controller for the motor position itself. The aerodynamic control loop is then wrapped around the motor controller. This is shown in the block diagram in Figure 6. The separation of these two controllers allows for different aerodynamic controllers to be designed that outputs the angle of attack based on the sensor feedback.

The motor controller receives a desired angle of attack (α_d) as input and commands the motor position (α). The motor position is tracked with an encoder which provides both position and velocity feedback. This is multiplied with gain values derived from poles of the motor plant and subtracted from the target position and velocity to arrive at the commanded values for the motor current. A system identification was performed on the motor plant with the wing attached at the freestream velocity of 0.371 m/s. This consisted of sinusoidal and step inputs to the motors current

command while reading the position and velocity information back from the encoder. This information was used to fit a transfer function and estimate the break frequencies and slopes of the system. The resulting transfer function was then converted to a state space representation with the values shown in Equation 4. This was then used to simulate the system and pick poles to achieve the desired performance of the system.

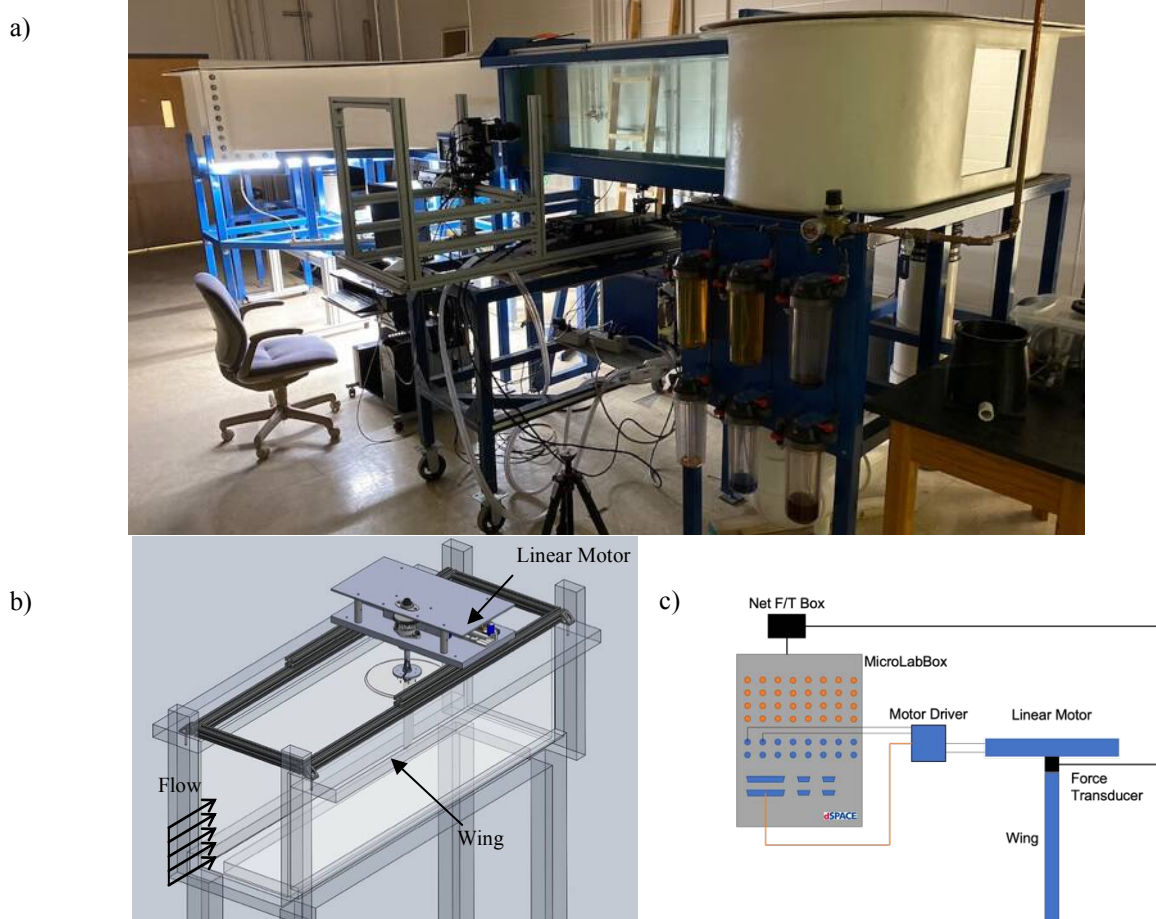


Figure 5 a) University of Dayton Water Tunnel facility b) CAD schematic of gust generator and test article c) wiring schematic of MicroLabBox and input/output devices

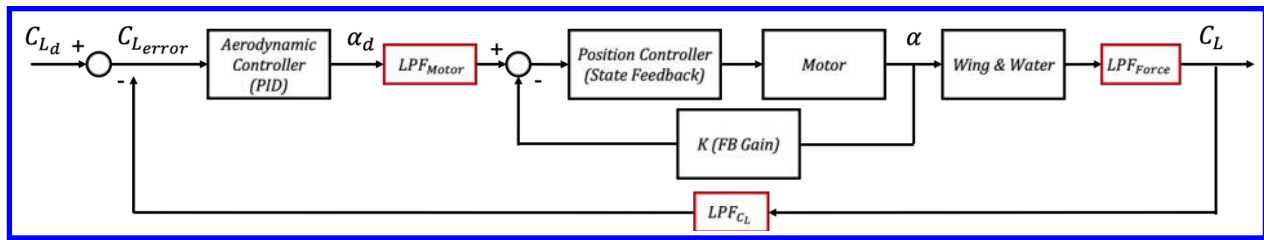


Figure 6 Block Diagram of System Containing Motor Controller and Lift Controller

The desired poles are listed in Equation 5. The resulting feedback gains of the system are shown in Equation 6.

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 170 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (4)$$

$$P = [-11 \quad -15.45] \quad (5)$$

$$K = [0.9997 \quad 0.1438] \quad (6)$$

Once the system identification and motor controller design were complete, position command of the motor was available and calibration of the motor position to the angle of attack of the wing was performed. This process included using a manual protractor on the shaft of the wing mount and moving the motor to offset the angle in increments of 45, 90, and 180 degrees. The starting and ending position values were recorded and used to compute an average displacement of 0.08356 centimeters per degree. This was used to convert inputs of the system from angle of attack in degrees to position for the motor controller. During the process of calibration, it was noticed that with inputs to the motor controller, step functions and other instantaneous inputs resulted in rapid acceleration and deceleration of the motor. It was decided that a transfer function filter would be used to filter higher frequency content from the input to avoid these rapid movements which might damage the sensor and wing once mounted and active. The low pass filter transfer function is shown in Equation 7.

$$LPF_{Motor} = LPF_{Forces} = \frac{1}{\frac{s}{100} + 1} \quad (7)$$

The closed-loop aerodynamic controller for the system is given a desired coefficient of lift input. The active CL is subtracted from the desired CL input and the result is used as the error input of a proportional, integral, derivative (PID) controller that outputs a new target or desired angle of attack of the wing in order to achieve the target CL. The active CL of the system is calculated by measuring the normal and axial forces using a Nano25 sensor. Since the sensor rotates with the wing, the active angle of attack of the system is used to compute the lift coefficient from the aerodynamic forces measured by the sensor. Filtering the force data is essential for successful operation of the control system. Without filtering, sensor noise and oscillations from the shedding of the wing produced large oscillations and disturbances in the CL rendering the control system ineffective. The same low pass filter transfer function as mentioned previously and shown in Figure 6. was used on both the normal and axial forces before CL calculation. After the CL was calculated, another transfer function filter was applied in order to further smooth out the oscillations of the signal. This transfer function is shown in Equation 8. It should be noted that this filter has a significant effect on the phase lag of the system since its denominator coefficient is 1/5.

$$LPF_{CL} = \frac{1}{\frac{s}{5} + 1} \quad (8)$$

The existence of large oscillations in the force data revealed the presence of 9Hz shedding behind the wing at low angles of attack. This was one of the primary reasons that the filtering described earlier was performed. The dye flow visualization shown in Figure 7 performed at a rate of 240 fps shows the existence of this shedding. Both the flow visualization and the time trace of forces confirmed the existence of vortex shedding at a rate of 9 Hz. It was also noticed that the tip of the wing deflected at a similar rate to the shedding frequency.

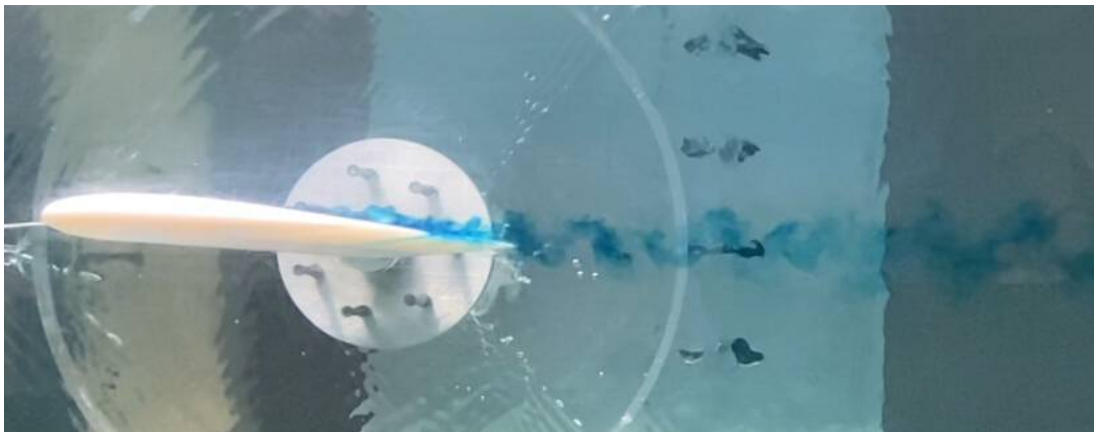


Figure 7 Dye Flow Visualization indicating vortex shedding in the wake of NACA 0015 wing at 0° angle of attack

III. Results

A. Closed loop control of static C_L

The initial tuning of the system was carried out by commanding a fixed angle of attack in an open loop controller in order to read the sensed forces and compute the resulting lift coefficient. This was done for angles of attack from -12 to 12 in increments of 1 degree. Figure 8 shows the resulting lift coefficient for each of these angles of attack in blue (labeled alpha hold). The closed loop controller was then used to command a fixed CL value from the system. CL values were commanded from -0.35 to 0.35 in increments of 0.05. The values for the closed loop command CL results are shown in red in Figure 8. Agreement of the closed loop CL commanded values to the open loop angle of attack commanded values established validity of the system to perform closed loop CL tracking for steady state aero results. It can be seen that the $C_{L_{Max}}$ of the wing is around 0.4. Therefore, the control system was never able to achieve a target CL beyond 0.4 and large divergence in angle of attack was seen.

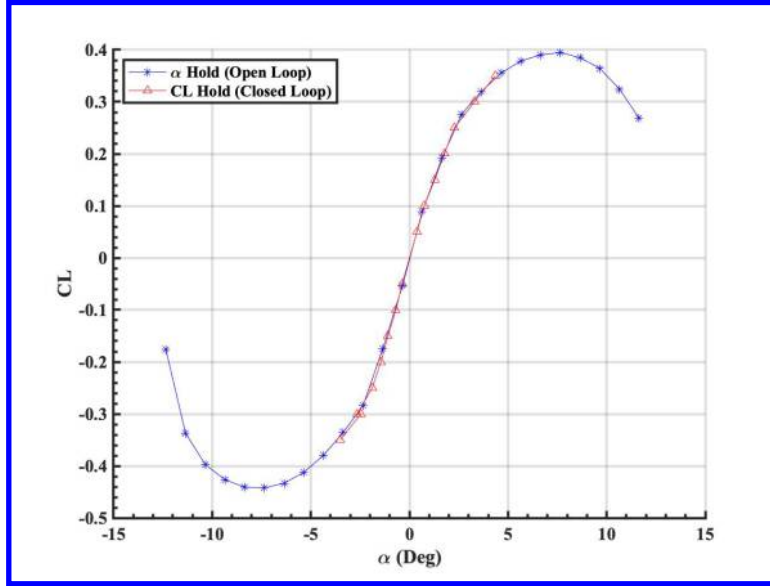


Figure 8 Static Lift Coefficient Comparison Between Open and Closed Loop Systems

B. Closed loop control of dynamic C_L

The efficacy of the control system to achieve a harmonic oscillation of lift coefficient was tested and the results are discussed in this section. The harmonic oscillation of the lift coefficient is given by,

$$C_{L_{Target}} = A_{Target} \sin(2\pi ft) \quad (9)$$

where the A is the amplitude and f is the frequency. The amplitude A was varied from $0.05 C_L$ to $0.4 C_L$ and the frequency was varied from 0.05 Hz to 0.35 Hz. Based on the freestream velocity and chord length of the wing, this resulted in a reduced frequency range between 0.04 to 0.3. The comparison of the input C_L and the C_L tracked by the closed loop control is shown in Figure 9 for a reduced frequency of 0.04, 0.12, and 0.25 to achieve a target amplitude C_L of 0.3. It can be seen that the amplitude of the C_L was matched well by the closed loop control system and as the reduced frequency increases, significant phase lag seem to occur between the target C_L and the C_L achieved by the closed loop control. The A_{Target} and the $A_{Closed Loop}$ is plotted against each other in Figure 10a to determine the reduced frequency at which $A_{Closed Loop}$ deviates from A_{Target} in Figure 9a. It can be seen that at lower reduced frequencies, the A_{Target} and $A_{Closed Loop}$ matches comparatively well. As the reduced frequency increases, the $A_{Closed Loop}$ exhibits a higher lift coefficient than the A_{Target} . This trend is once again seen in Figure 10b where closed loop C_L amplitude is normalized with the target C_L amplitude and is shown as a function of reduced frequency. An ideal case would be to have all the curves stay at a ratio of 1 across all reduced frequencies.

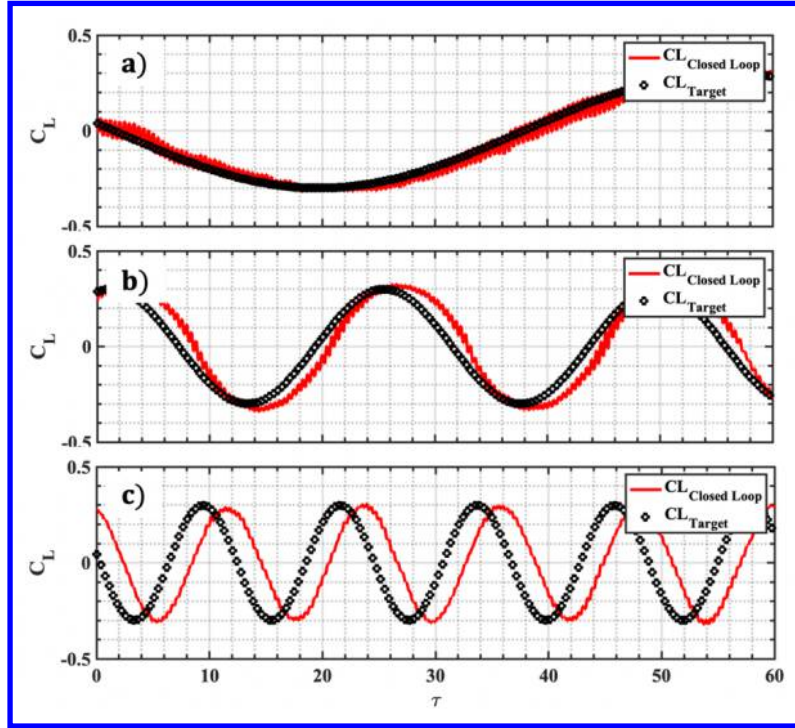


Figure 9 Comparison of target C_L variation with the variation achieved by closed loop control at a) $k = 0.04$, b) $k = 0.12$, and c) $k = 0.25$

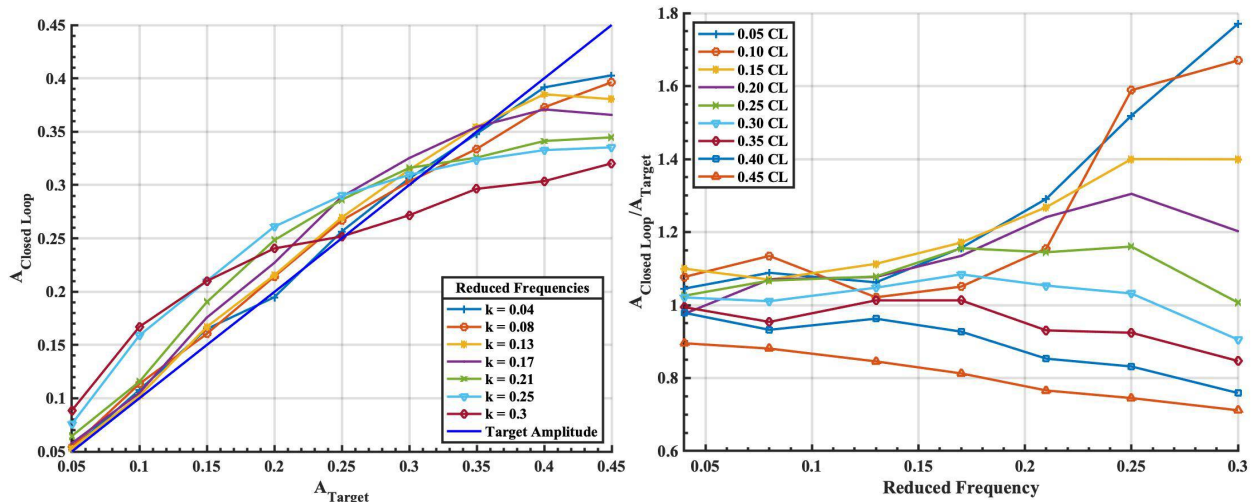


Figure 10 Closed Loop Sensed Coefficient of Lift Amplitude Versus Target Amplitude

At lower reduced frequencies, the ratio of the target and closed loop control C_L is close to 1. However with increment in reduced frequencies, it can be seen that the $A_{Closed Loop}$ diverges from A_{Target} . At lower target amplitudes, increase in the reduced frequencies causes the closed loop control C_L to overshoot the target amplitude by almost 60% at a reduced frequency of 0.3. However, at moderate target amplitudes such as a C_L of 0.3, the ratio of the closed loop and target amplitude is roughly around 1. This is once again evidenced by the lift coefficient histories shown in Figure 9. At amplitudes higher than 0.3, increment in reduced frequency lowers the amplitude achieved by the closed loop control with maximum decrement of around 20% at a reduced frequency of 0.3. The changes in amplitude is also accompanied by changes in the phase angle between the target and the closed loop control lift coefficient history as shown in Figure 11. In Figure 11 it can be observed that as the reduced frequency of the system increases, the phase lag of the CL response increased for almost all of the cases. This phase lag is currently hypothesized to be due to the

filtering of the high frequency signals as shown in Figure 6. Even at lower reduced frequencies, the phase lag increases with increase in the target C_L amplitude. However, the phase lag almost linearly increases with increase in reduced frequency. With better tuning of the PID controller at different reduced frequencies, the difference between the target coefficient of lift amplitude and the amplitude achieved by closed loop control can be minimized along with minimizing the associated phase lag. This effort is currently being undertaken and will be presented in the follow-up paper.

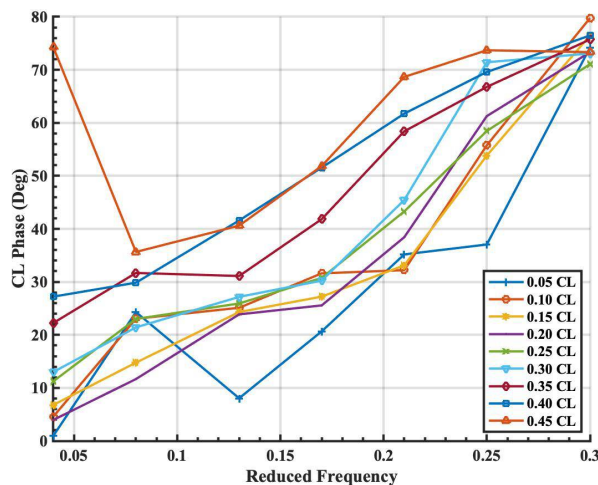


Figure 11 Phase lag variation between the target coefficient of lift history and closed loop control history as a function of reduced frequency

C) Comparison with Theodorsen's model

The angle of attack variation obtained from the closed loop control system to achieve a desired coefficient of lift history is compared with the angle of attack variation required by the Theodorsen's model to achieve the same target coefficient of lift history. This comparison is shown in Figure 12 for the lowest (0.04) and highest (0.3) reduced frequency cases with each plot containing the comparison of the angle of attack history of the system with the expected history from Theodorsen's, the target CL from our sine generator aligned with the Theodorsen model CL compared with the sensed CL of the system. It is apparent in low reduced frequency cases, the ability of the controller to track the target is much better than at higher reduced frequencies. The agreement between the angle of attack variation from the closed loop control agrees extremely well with the Theodorsen model's prediction. However, at higher reduced frequencies, as shown in the previous section, the target coefficient of lift is not achieved by the closed loop control system. Significant differences were observed in the C_L amplitude and there is an apparent phase lag as well. Therefore, the corresponding angle of attack variations didn't agree well that of the Theodorsen model.

The phase lag between the angle of attack time trace obtained from Theodorsen model and the control system is shown in Figure 13. It is interesting to see that the trends appear to be more dependent on the commanded CL amplitude and generally maintain similar phase throughout the range of reduced frequencies. It can also be seen that at lower CL amplitudes the phase of the system is negative, performing the angle of attack maneuvering before the Theodorsen model prediction whereas at CL amplitudes of 0.3 and above the system response is behind the Theodorsen model prediction.

In the angle of attack amplitude plot shown in Figure 14 it can be seen that at lower reduced frequencies the system produces higher angles of attack than Theodorsen predicts for the higher CL amplitude cases while at lower CL amplitudes for lower reduced frequencies the system aligns well with the Theodorsen models predicted angle of attack amplitudes. An interesting trend apparent in this plot is the crossover between reduced frequencies of 0.2 and 0.25, it appears that as a case such as the 0.45 CL amplitude is increased in reduced frequency, the alignment with the Theodorsen model angle of attack is improved, potentially to a much better match if tested beyond the frequencies used in this study. Looking at the context these plots present together its apparent that the 0.3 CL amplitude case provides the best alignment to the Theodorsen model for both angle of attack phase and amplitude for the range of reduced frequencies tested.

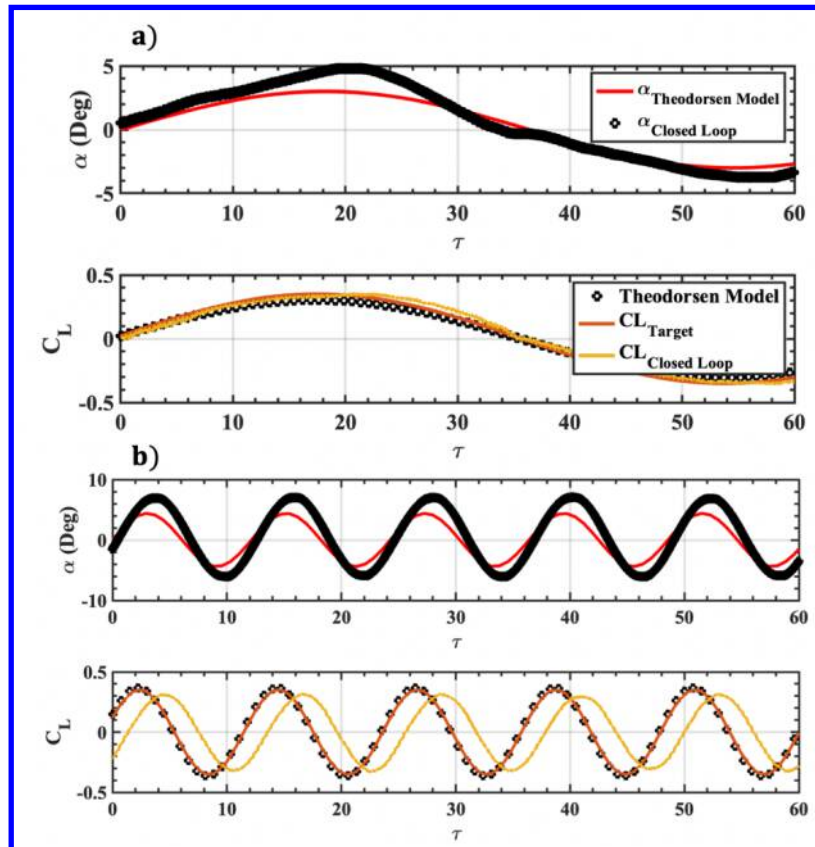


Figure 12 Comparison of angle of attack variation between the closed loop control and the Theodorsen model at a) reduced frequency of 0.04 and b) reduced frequency of 0.3

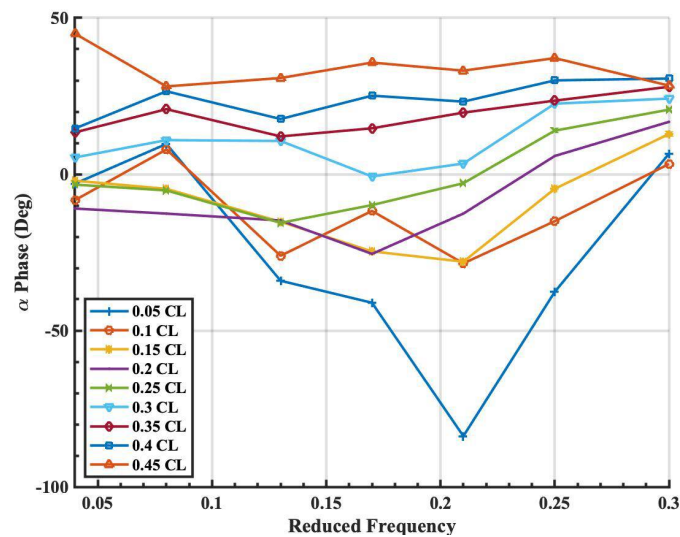


Figure 13 Phase of Closed Loop System to Theodorsen Angle of Attack Versus Reduced Frequency

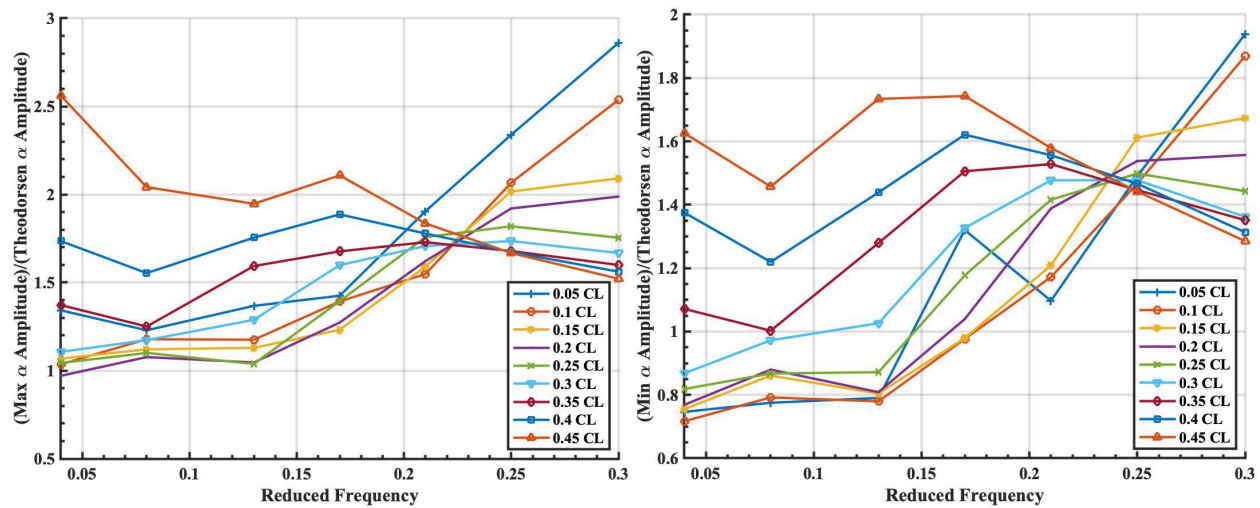


Figure 14 Amplitude of Closed Loop System to Theodorsen Angle of Attack Versus Reduced Frequency

IV. Conclusions

A closed-loop feedback control system was designed and built using a force transducer on an AR 2.5 wing to perform active CL tracking with the intent of developing a system for gust mitigation. This system comprised of a motor controller that implemented a state feedback control strategy that accurately tracked the motor position to the desired position of the system. Wrapped around the motor controller was an aerodynamic PID controller implementing a closed-loop control strategy to track a desired coefficient of lift. The aerodynamic controller failed to track accurately in all of the cases tested but proved accurate in lower reduced frequency cases with lower CL amplitudes. Moreover, the results showed that the motor controllers' ability to accurately track position would be suitable for further studies implementing different aerodynamic control strategies.

Further development of the aerodynamic controller and implementation of more robust control architectures would yield more accurate results for tracking of a desired coefficient of lift. The implementation of a state feedback based aerodynamic controller along with the addition of relevant states into the controller would likely improve the CL tracking of the system.

V. References

- [1] Andreu Angulo, Ignacio, et al. "Wing-gust interactions: The effect of transverse velocity profile." *AIAA Scitech 2020 Forum*. 2020.
- [2] Sedky, Girguis, Anya R. Jones, and Francis D. Lagor. "Lift Regulation During Transverse Gust Encounters Using a Modified Goman-Khrabrov Model." *AIAA Journal* (2020): 1-11.
- [3] Yeo, Derrick, Vikram Hrishikeshavan, and Inderjit Chopra. "Gust Detection and Mitigation on a Quad Rotor Biplane." In *AIAA Atmospheric Flight Mechanics Conference*, p. 1531. 2016.
- [4] Praveen, Viswanadhapalli, and S. Pillai. "Modeling and simulation of quadcopter using PID controller." *International Journal of Control Theory and Applications* 9, no. 15 (2016): 7151-7158.
- [5] Murua, Joseba, Rafael Palacios, and J. M. R. Graham. "Open-loop stability and closed-loop gust alleviation on flexible aircraft including wake modeling." In *53rd AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Materials Conference 20th AIAA/ASME/AHS Adaptive Structures Conference 14th AIAA*, p. 1484. 2012.
- [6] Asgharnia, A., A. Jamali, R. Shahnazi, and A. Maheri. "Load mitigation of a class of 5-MW wind turbine with RBF neural network based fractional-order PID controller." *ISA transactions* 96 (2020): 272-286.
- [7] Kerstens, W., Pfeiffer, J., Williams, D., King, R. and Colonius, T., 2011. Closed-loop control of lift for longitudinal gust suppression at low Reynolds numbers. *AIAA journal*, 49(8), pp.1721-1728.
- [8] Leishman, Gordon J. *Principles of helicopter aerodynamics with CD extra*. Cambridge university press, 2006.